THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT-1
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LAB -1

Topic Covered - cin/cout basics, structures in C, structures in C++, Classes and objects

Programming Question

1. Write a program (WAP) to display "Hello World" on console display.

bitwise_swapping.cpp library charges.cpp igor.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    cout<<"hello world";
}

```

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```
hello world
--------------------------------
Process exited after 0.2325 seconds with return value 0
Press any key to continue . . .
```

2. WAP to that reads two values from keyboard and displays the larger value on the screen.

bitwise_swapping.cpp | libracy charges.cpp | rgbF.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
  int a;
  cin>>a;
  cout<<"the value of a is"<<a;
  int b;
  cin>>b;
  cout<<"\nthe value of b is :"<<b;
  if(a>b){
    cout<<"\nthe value of the largest value is:"<<a;
  }
  else{
    cout<<"the value of the largest value is:"<<b;
  }
}
```

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```
20
the value of a is20
10

the value of b is :10
the value of the largest value is:20
--------------------------------
Process exited after 5.25 seconds with return value 0
Press any key to continue . . .
```

3. WAP to read three values a, b, c from keyboard and display the value of x = a/b-c.

bitwise_swapping.cpp | libracy charges.cpp | rgbF.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
  int a;
  cin>>a;
  cout<<"the value of a is"<<a;
  int b;
  cin>>b;
  cout<<"\nthe value of b is :"<<b;
 int c;
 cin>>c;
 cout<<"\nthe value of c is:"<<c;
 int x;
 x=a/(b)-c;
 cout<<"\nthe value of x is :"<<x;
}
```

```
10
the value of a is10
3

the value of b is :3
4

the value of c is:4
the value of x is :-1
--------------------------------
Process exited after 7.338 seconds with return value 0
Press any key to continue . . .
```

4. WAP to implement the following control characters:
   - *'\n' is for new line, or you can use endl – cout<<endl<<"message";*
   - *'\t' is for tab*
   - *'\a' is an alarm sound*
   - *'\r' is carriage return to go to the beginning of the current line*

bitwise_swapping.cpp | libracy charges.cpp | rgbF.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    cout<<"my name is \n rohan";
    cout<<"my name is\t rohan";
    cout<<"my name is rohan\a";
    cout<<"\a alarm sound";
    cout<<"\ryo";
}

```

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```
my name is
yoohanmy name is         rohanmy name is rohan alarm sound
--------------------------------
Process exited after 0.1837 seconds with return value 0
Press any key to continue . . .
```

5. Implement *namespace* in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

bitwise_swapping.cpp | libracy charges.cpp | rgbF.cpp

```cpp
#include <iostream>
using namespace std;
namespace A{
    int a=5;
     void display(){
     cout<<"the value of a is:"<<a;

     }

}
namespace B
{
int b=10;
    void display()
    {
    cout<<"the value of b is:"<<b;


    }
}
int main() {
A::display();//function call
B::display();

cout<<"the value of A is "<<A::a;
}
```

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```
the value of a is:5the value of b is:10the value of A is 5
--------------------------------
Process exited after 0.1658 seconds with return value 0
Press any key to continue . . .
```

6. Write a program to define a structure *Student* that contains 3 variables and 2 functions:
   - *Variables:*
     (a) name
     (b) rollget
     (c) marks
   - *Functions:*
     (1) setStudentData()
     (2) getStudentData()

Use loops to input and output data for 3 student object entries.

```cpp
#include<iostream>
using namespace std;

    struct student{
        private:
        char name[20];
        int roll_no;
        int marks;
    public:
        void setStudentData(){//no display of the names ,roll no ,marks as its set


            cout<<"the name of the student is:";
        cin>>name;

            cout<<"\nthe roll no of the student is:";
            cin>>roll_no;


            cout<<"\nthe marks of the student is :";

    cin>>marks;
        }

        void getStudenData(){//display of student data as its get
            cout<<"\nname of the studnet is"<<name;
            cout<<"\nthe roll no of the student is"<<roll_no;
            cout<<"\nthe marks of the student is;"<<marks;
        }

    };
    int main(){
        student s[3];
    for(int i=1;i<=3;i++)
{
    cout<<"\nenter the detial of student :\n"<<i;
    s[i].setStudentData();

    }
```

```
enter the detial of student :
1the name of the student is:rohan

the roll no of the student is:187

the marks of the student is :20

enter the detial of student :
2the name of the student is:rihan

the roll no of the student is:188

the marks of the student is :30

enter the detial of student :
3the name of the student is:johan

the roll no of the student is:189

the marks of the student is :40

--------------------------------
Process exited after 38.57 seconds with return value 3221225477
Press any key to continue . . .
```

7. In the previous program make the data private, and functions public.

function.cpp | structure and function.cpp

```cpp
#include<iostream>
using namespace std;

    struct student{
        char name[20];
        int roll_no;
        int marks;
        void setStudentData(){//no display of the names ,roll no ,marks as its set


            cout<<"the name of the student is:";
        cin>>name;

            cout<<"\nthe roll no of the student is:";
            cin>>roll_no;


            cout<<"\nthe marks of the student is :";

    cin>>marks;
        }

        void getStudenData(){//display of student data as its get
            cout<<"\nname of the studnet is"<<name;
            cout<<"\nthe roll no of the student is"<<roll_no;
            cout<<"\nthe marks of the student is;"<<marks;
        }

    };
    int main(){
        student s[3];
    for(int i=1;i<=3;i++)
{
    cout<<"\nenter the detial of student :\n"<<i;
    s[i].setStudentData();

    }
```

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```
enter the detial of student :
1the name of the student is:rohan

the roll no of the student is:187

the marks of the student is :44

enter the detial of student :
2the name of the student is:ayush

the roll no of the student is:221

the marks of the student is :44

enter the detial of student :
3the name of the student is:divyam

the roll no of the student is:44

the marks of the student is :45

--------------------------------
Process exited after 42.19 seconds with return value 3221225477
Press any key to continue . . .
```

8. Convert program #4 from struct to class.

function.cpp | structure and function.cpp | WELL DONE.cpp

```cpp
#include<iostream>
using namespace std;

    class student{
        private:
        char name[20];
        int roll_no;
        int marks;
    public:
        void setStudentData(){//no display of the names ,roll no ,marks as its set


            cout<<"the name of the student is:";
        cin>>name;

            cout<<"\nthe roll no of the student is:";
            cin>>roll_no;


            cout<<"\nthe marks of the student is :";

    cin>>marks;
        }

        void getStudenData(){//display of student data as its get
            cout<<"\nname of the studnet is"<<name;
            cout<<"\nthe roll no of the student is"<<roll_no;
            cout<<"\nthe marks of the student is;"<<marks;
        }

    };
    int main(){
        student s[3];
    for(int i=1;i<=3;i++)
{
    cout<<"\nenter the detial of student :\n"<<i;
    s[i].setStudentData();

```

```
enter the detial of student :
1the name of the student is:rohna

the roll no of the student is:33

the marks of the student is :33

enter the detial of student :
2the name of the student is:tom

the roll no of the student is:22

the marks of the student is :11

enter the detial of student :
3the name of the student is:harry

the roll no of the student is:22

the marks of the student is :77

--------------------------------
Process exited after 42.01 seconds with return value 3221225477
Press any key to continue . . .
```

9. Discuss the differences of structures in C and structures in C++ during the lab with other students or your instructor.

Sr.	C language	C++
1	Functions are basic elements	Classes are basic elements
2	Focus on global functions	Focus on encapsulation (data+functions)
3	Functions share global data	Data and function access is controlled
4	Data moves openly	Data is bounded with classes
5	Top down approach (break down into functions)	Bottom up (clubbing of data and functions together)

10. WAP to read an integer value from the keyboard and display on screen "WELL DONE" that many times.

function.cpp [*] structure and function.cpp WELL DONE.cpp

```cpp
#include<iostream>
using namespace std;
int main(){
    int a;
    cin>>a;
    cout<<"the value of a is:"<<a;
    for(int i=0;i<=a;i++)
{
    cout<<"\nWELL DONE\n";
}

}
```

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```
10
the value of a is:10
WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

WELL DONE

--------------------------------
Process exited after 1.702 seconds with return
Press any key to continue . . .
```